# Matthew Pidden

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## Education

**Imperial College London** – MSc Visual Computing and Robotics, Distinction (expected) 2025 – 2026
- **Award:** Best Coursework in Robot Learning (won class-wide knockout competition between learned robot policies)
- **Coursework:** Robotics, Robot Learning, Reinforcement Learning, Machine Learning, Software Engineering for ML Systems, Scalable Systems and Data

**University of Bristol** – BSc Computer Science, First Class Honors 2022 – 2025
- **Awards:** Hargrave Prize (best final-year project), Netcraft Prize (top 10 in year), Best Software Engineering Group Project. Lead Teaching Assistant (2024 – 25).
- **Coursework:** Machine Learning, Computer Vision, Artificial Intelligence, High Performance Computing, Concurrent and Distributed Programming

## Experience

**ActionQuest** – Program Director & Skipper, BVI & Leeward Islands Summers 2022 – 2025
- Skippered 50 ft yachts with 2 crew and 12 students for 3 week sessions; ran passage planning, navigation, repairs, rigging and electrical faults with limited shore support.
- Program Director (2024 & 2025): led a 5 vessel fleet of 16 staff and 60 students, owning scheduling, safety, communications, repairs, emergency response and staff management. Most senior on-site leader for international passages, handling fuel, water, provisioning, and customs and immigration across 6 countries.

**Global Expeditions Group** – Software Engineer, Remote Oct 2023 – May 2025
- Built and deployed two offline-first, cross-platform Flutter apps (blog management and a staff portal) for 20 vessels in remote locations with intermittent connectivity.

**Clipper Round the World Yacht Race** – Second Mate, North Pacific Mar – Apr 2022
- Called up at short notice to join a team for a 36 day North Pacific crossing, leading watches and responsible for crew safety, navigation, and leadership alongside the skipper and first mate.

## Projects

**Autonomous Sailing USV (in progress)** – *Fusion 360*, *3D Printing*, *Raspberry Pi*, *ROS2*, *Python*, *Electronics* Present
- Designing and building a low cost 0.75 m autonomous sailing vessel for coastal data collection, with a 3D printed hull, keel, rudder and wing sail designed in Fusion 360.
- Integrating an ultrasonic anemometer, 9-axis IMU, environmental sensors, camera, GPS, cellar communication on a Raspberry Pi running ROS2 and a Pico on MicroPython.

**End-Effector Trace Conditioning for VLAs (MSc Thesis)** – *PyTorch*, *LeRobot*, *SO-101*, *Slurm*, *HPC* 2026
- Proposed EETC, a method that improves task generalisation in Vision-Language-Action models without architectural changes or new data requirements: 17.5% increase success rate on unseen tasks on real world hardware.
- Trained and fine-tuned models on the ISAMBARD-AI supercomputer (GH200) and local Blackwell GPUs.

**The Revenger – Guinness World Record (BSc Thesis)** – *C++*, *Python*, *Arduino*, *PCB Design*, *OpenCV*, *Fusion 360* 2025
- Built a 10 stepper-motor electromechanical system with a custom PCB, embedded firmware, novel algorithmic solver, and OpenCV vision pipeline for real-time state detection.
- Fastest robot to solve a 4x4 Rubik's Cube: 45.3 s, beating a record that stood for over a decade (previous 1:18.7). Covered by National TV and Radio News.

**Consumer Apps** – *Flutter*, *Firebase*, *RevenueCat*, *AdMob* 2017 – Present
- Built and marketed Santa Tracker: 90k downloads, #8 overall on the UK App Store, $8.3k revenue from 1,300 purchases and 390k ad impressions.

## Skills

**Robotics & Mechatronics**: ROS 2, robotic systems integration, electromechanical systems, embedded systems, sensor integration, actuator control, motor control, firmware development, Raspberry Pi, Arduino
**Mechanical Design**: Fusion 360, 3D CAD, mechanical prototyping, PCB Design, 3D printing, soldering
**Software:** Python, PyTorch, Hugging Face, LeRobot, OpenCV, computer vision, vision-language-action (VLA) models, reinforcement learning, Slurm/HPC, C++, C, Python, Go, Dart, JavaScript, Git, CI/CD, REST APIs, Flutter, Firebase, AWS
**Maritime & Field Operations**: IYT Yachtmaster Offshore, 20,000+ offshore miles, vessel maintenance